

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459553.9347 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.014
Transit depth (Rp/Rs)^2	2.64 +/- 0.47 [%]
Orbital Inclination (inc)	83.59 +/- 1.67
Airmass coefficient 1 (a1)	0.893 +/- 0.01
Airmass coefficient 2 (a2)	0.0535 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	#3 - [130, 208]
Optimal Aperture	3.69
Optimal Annulus	17.07
Transit Duration (day)	0.0547 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.014 vs 0.1594 (deviation 0.26σ)
Duration fit vs published	0.0547 d vs 0.0512 d (deviation 0.77σ)
Mid-time O–C	-5.3 min
Depth SNR	11.6
Residual scatter	1.17 %

Light curve

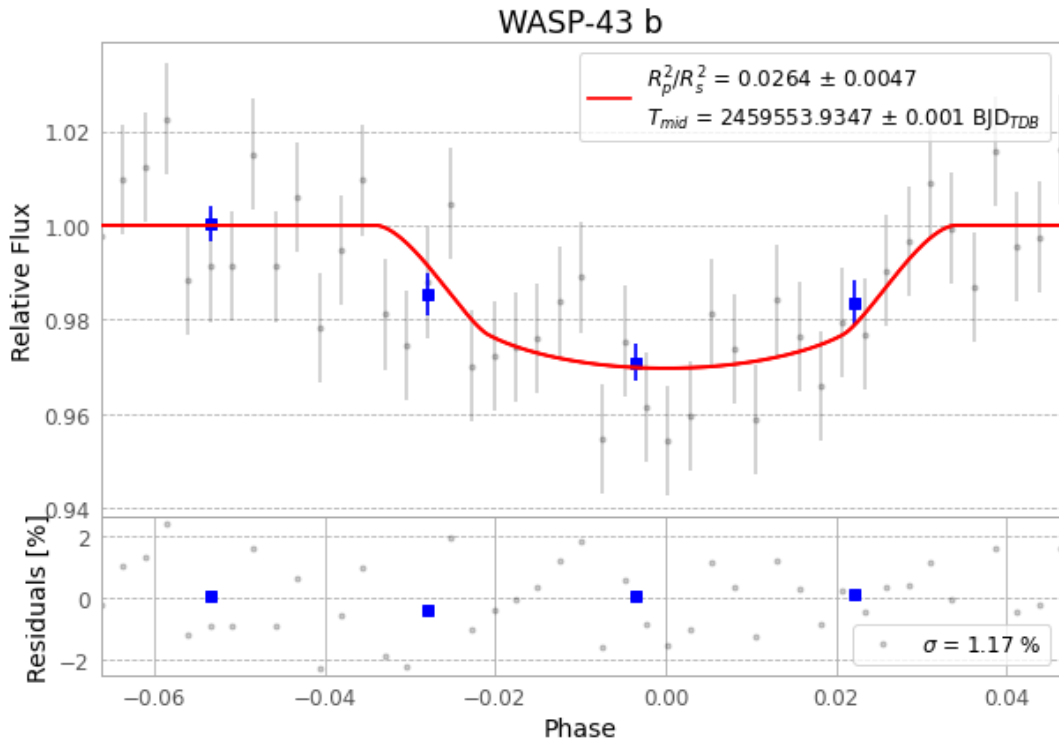


Figure 1 — FinalLightCurve_WASP-43 b_2021-12-05.png (SHA-256 bdba262cb319cf15...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 145], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fbc82c71882e63d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

45 FITS frames (30 MB), WASP-43211205090612.FITS → WASP-43211205111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-211205.log (a3d71767fc7b...), FinalLightCurve_WASP-43 b_2021-12-05.png (bdba262cb319...), FinalLightCurve_WASP-43 b_2021-12-05.csv (0870bbdeac1f...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:49Z.